V-M-81F
April 22, 1975
SUPERSEDING
Fed. Spec. V-M-81E
January 15, 1969
(See 6.6)

FEDERAL SPECIFICATION

MATTRESSES, BED, FELT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for taped box edge laminated felt mattresses.

1.2 Classification.

1.2.1 Grades and sizes. The mattresses shall be of the following grades and sizes as specified (see 6.2):

Grade A - 100 percent cotton
Grade B - 25 percent cotton, 75 percent linters
Grade C - 40 percent cotton, 60 percent linters
Grade D - 65 percent polyester, 35 percent linters

1.2.2 Sizes. Sizes of mattresses shall be as shown in table I, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276	-	Thread, Cotton
V-T-285	-	Thread, Polyester
CCC-C-436	-	Cloth, Ticking Twill, Cotton
DDD-L-20	-	Label; For Clothing, Equipage, and Tentage, (General Use)

FSC 7210

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Federal Standards:

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies)
FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-M-3996	-	Mattresses (Except Pneumatic Type), Packaging and Packing Of
MIL-B-41826	-	Batting, Synthetic Fibers: Polyester, (Quilted and Unquilted)

Military Standard

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

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U. S. Department of Agriculture Publication

C&MS-SRA-180 - Cotton and Cotton Linters Standards

(Application for copies should be addressed to the Department of Agriculture, Washington, DC 20250.)

Consumer Product Safety Commission

FF-4-72 - Flammability Standard For Mattress

(Application for copies should be addressed to the Consumer Product Safety Commission, 7315 Wisconsin Avenue, (East Tower) Bethesda, MD 20207.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.3 and 6.3).

3.2 Flammability. All mattresses shall meet the provisions of the Consumer Product Safety Commission Mattress Flammability Standard FF-4-72.

3.3 Materials.

3.3.1 Felt. The felt shall be clean and well garnetted to produce unbroken laminated sheets. The use of oil in the fabrication of the felt shall not be permitted.

3.3.1.1 Grade A. Grade A felt shall be made of 100 percent cotton conforming to 3.3.1.5.

3.3.1.2 Grade B. Grade B felt shall be made of a blend of 25 percent cotton conforming to 3.3.1.5 and 75 percent linters conforming to 3.3.1.6.

3.3.1.3 Grade C. Grade C felt shall be made of a blend of 40 percent cotton conforming to 3.3.1.5 and 60 percent linters conforming to 3.3.1.6.

3.3.1.4 Grade D. When specified (see 6.2), grade D felt shall be made from a blend of 65 percent polyester fiber conforming to 3.3.1.7 and 35 percent linters conforming to 3.3.1.6.

3.3.1.5 Cotton. The cotton shall be new (not previously used) minimum, 78 inch staple cotton and not lower in grade than the lowest grade in the white, spotted, or gray grades, conforming to the standard of the United States Department of

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Agriculture Publication No. C&MS-SRA-180 (see 2.2). The use of reginned cotton will be permitted provided the required cotton is equivalent in grade to the standard listed above. Tinged or stained cotton shall not be permitted. Testing shall be as specified in 4.4.1.1.

3.3.1.6 Linters. The linters for grades B, C, and D mattresses shall be new (not previously used), conforming to grade 5, staple 5, of USDA Publication No. C&MS-SRA-180.

3.3.1.7 Polyester fiber. The fiber to be blended with the linters specified in 3.3.1.6 for grade D felt (see 3.3.1.4) shall consist of a minimum range of 20 to 30 percent mixture of crimped, (6 to 8 crimps per inch) cut staple virgin polyester fiber, of 5.0 or 6.0 denier, 2 ± 1/2 inch in length and the balance of the mixture shall be made from any polyester fiber (95 percent pure polyester fiber). Testing shall be as specified in 4.4.1.2.

3.3.2 Ticking. The cotton ticking for the mattress case shall conform to type I or type II, class 1 or 2 of CCC-C-436. When fire-resistant ticking is required (see 6.2), the treated ticking shall conform to type II, class 2 of CCC-C-436.

3.3.3 Thread. The thread used for stitching the edge binding tape and the border shall be cotton, ticket number 30, 3 ply, type IA1 or IA2 of V-T-276; or the thread shall be polyester type I, class 1, sub-class A, size E of V-T-285.

3.3.4 Tape.

3.3.4.1 Edge binding tape. The tape used in finishing the edges of the mattress shall be 5/8 inch (minimum) width polyester tape or an approved fire-resistant or smoldering ignition-resistant treated cotton tape (see 6.4). The polyester tape shall have a minimum breaking strength of 100 pounds and the treated cotton tape shall have a minimum breaking strength of 30 pounds. The color of the tape shall be white or natural. Testing shall be as specified in 4.4.1.1.

3.3.4.2 Tufting tape, braid or twine. The tape, braid or twine used for tufting shall be either nylon or polyester. The nylon or polyester tape shall be not more than 5/8 inch wide and shall have a 50 pound minimum breaking strength. The breaking strength for the nylon or polyester braid or twine shall be 80 pounds minimum. Innertufted mattresses can use cotton tape, braid or twine for tufting. The cotton tape shall be not less than 3/8 inch nor more than 5/8 inches wide and shall have a minimum breaking strength of 50 pounds. The cotton braid or twine shall have a breaking strength of not less than 32 pounds. Testing shall be as specified in 4.4.1.1.

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3.3.5 Barrier materials.

3.3.5.1 Polyester batting. Polyester batting conforming to type III batting of MIL-B-41826 may be used as a barrier, except that the requirement for thickness after laundering, compressional recovery after laundering, and launderability need not apply.

3.3.5.2 Commercial bonded polyester batting. The use of commercial bonded polyester batting is acceptable provided that the batting is bonded with an approved durable bonding agent (see 6.5) uniformly applied. The maximum concentration of the applied bonding agent shall not exceed 18.0 percent on the weight of the finished bonded batting. The bonding agent shall be applied to both sides of the batting filler.

3.3.5.3 Approval and qualification of other barrier materials. Other commercial mattress barrier materials or approved smoldering-resistant treatments may also be used as barrier materials. Approval of these materials and or treatments must be obtained prior to use (see 6.4). The use of an impermeable material for use as a barrier is prohibited (see 4.4.1.2).

3.3.6 Open mesh or nonwoven fabric. The open mesh fabric shall have a weight of 1-1/2 ounces to 3-1/2 ounces per square yard and shall be a thermoplastic material. The spun bonded nylon, polyester or polypropylene nonwoven fabric shall have a weight of 1-1/2 to 3-1/2 ounces per square yard (see 3.5.10.1 and 4.4.1.2).

3.3.7 Heat perimeter shield or flange. The material used for flanging or a heat perimeter shield shall be an aluminized fabric (see 4.5.1.2).

3.3.8 Labels. All labels shall be sewn at one end of the mattress into the same edge seam.

3.3.8.1 Statutory label. Statutory labels shall conform to the legal requirements of the state in which the mattresses are manufactured.

3.3.8.2 Identification label. Each mattress shall have an identification label conforming to type I, class 5 of DDD-L-20, except that the requirement for launderability shall not apply. The size of characters of the inscription on the label shall be 3/32 inch to 1/4 inch. The flammability lot identification number shall be included in the marking.

3.3.8.3 Medical label. When specified (see 6.2), mattresses procured for medical activities shall have a label conforming to type I, class 9 of DDD-L-20, bearing manufacturer's name or registered trade mark, FSN, item description and the flammability lot identification number.

3.4 Design. Mattresses shall be either outertufted or innertufted. The detailed construction requirements shall be as specified in 3.5.

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3.5 Construction. The construction requirements for the mattresses shall be as specified herein.

3.5.1 Stitching. Stitching shall conform to stitch type 301 or 401 of FED-STD-751 with 8 to 12 stitches per inch. When 401 stitching is used, the chain portion of the stitching shall not appear on the outside of the ticking.

3.5.2 Seams. When longitudinal pieces of ticking are used they shall be stitched with stitch type 401 and seamed with seam type LSb-2 or stitch type 301 and seam type SSa-1 conforming to FED-STD-751.

3.5.3 Type 301 stitching. Ends of all seams and stitching when not caught in other seams and stitching shall be backstitched or overstitched 3/4 inch minimum. Backstitching or overstitching is not required when closing the mattress on a tape edge machine. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.5.3.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 3/4 inch back of the end of the stitching. 1/

b. Thread breaks, or two or more consecutive skipped stitches or run-offs noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 3/4 inch in back of the defective area, continue over the defective area and continue a minimum of 3/4 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

3.5.4 Types 401, 504 and 505. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.3.1(a) and 3.5.3.1(b), and type 301 stitching may be used for making the repairs of type 401 stitching.

3.5.5 Cover construction. The top and bottom sections in the mattress cover shall each be made up of not more than two longitudinal pieces of ticking specified in 3.3.2. The ticking shall be matched and sewn together (see 3.5.2). The blue

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stripes of the ticking shall run the lengthwise direction of the mattress. The top and bottom panels shall have hemmed edges when flanging or a heat perimeter shield is not used.

3.5.5.1 Flange or heat perimeter shield attachment. When a flange or heat perimeter shield is used, it shall be 1-3/8 inches to 5-1/2 inches wide and may be an impermeable material such as aluminized fabric (see 3.3.7). The flange or heat perimeter shield shall be attached to the ticking by using one of the following operations:

(a) By overedge stitching flange or heat perimeter shield to the raw edge of the ticking.

or

(b) By superimposing the flange or heat perimeter shield on the ticking and seaming with a row of stitches 1/16 to 1/8 inches from the edge.

or

(c) By preheming the ticking and seaming the flange or heat perimeter shield on the ticking with a row of stitches 1/16 to 1/8 inches from the edge or the operation may be accomplished simultaneously.

The flange or heat perimeter shield shall also be stitched to the ticking with a second row of seaming stitches (stitch type 301 or 401). This row of stitching shall be positioned along the opposite long edge of the flange or heat perimeter shield material and shall be sewn a minimum of 1/4 inch from the edge. Both rows of stitching of the heat perimeter shield or flange, after overstitching or back-stitching the ends of stitching, may be carried off the side edge of the ticking cover and then trimmed. When 401 stitching is used, the chain portion of the stitching shall not appear on the outside of the ticking. When overedge stitching is used it shall be accomplished with stitch type 504 or 505, 3/16 inch gauge of throw, 5 to 7 stitches per inch. The second row of seaming stitches shall have 5 to 7 stitches per inch. Thread shall be as specified in 3.3.3.

3.5.6 Boxing and corners. The boxside panel of the mattress shall be formed of not more than four separate pieces of ticking specified in 3.3.2. When more than one piece of ticking is used in constructing the boxside panel, the pieces shall be of sufficient length to allow joining of seams to be at a corner of the mattress and the pattern shall be properly matched. The corners of the mattress shall be slightly rounded to permit the use of the tape edge machine.

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3.5.7 Edge construction. The finished edge of the cover ticking and the finished edge of the boxside panel shall be joined together and sewn within the folds of the edge binding tape specified in 3.3.4.1. The tape shall extend completely around both edges of the mattress. Not more than three pieces of tape shall be used on each border edge.

3.5.8 Felt distribution. From 80 to 90 percent of the felt shall be distributed in layers of unbroken laminated felt sheets of uniform weight and thickness. Each side of the mattress shall have a special crown of felt located over the middle one-third of the mattress and across the full width of the mattress. The crowns shall consist of 10 to 20 percent of the total weight of the felt.

3.5.9 Barrier material (toppers). The supplier shall use a barrier system or an approved flame retardant or smoldering ignition-resistant treatment to meet the mattress flammability provisions specified in 3.2. When a barrier system is used, the barrier materials specified in 3.3.5 shall be placed directly over the felt padding. Innertufted mattresses can require use of a tufting support fabric (see 3.3.6) over the barrier materials. The use of a tufting support fabric (see 3.3.6) for outer tufted mattresses is optional. When approved treatments are to be applied to the felt padding the use of a barrier system or tufting support fabric is optional.

3.5.10 Tufting. Mattresses may be designed to be either outertufted or innertufted. Tufting may be accomplished using either tufting twine, braid or tape specified in 3.3.4.2. The depth of tufts shall be 2 inches to 3-1/2 inches in length for mattresses having a thickness of 4-1/2 inches or less. The depth of tufts for mattresses over 4-1/2 inches in thickness shall be 4-1/2 to 5-1/2 inches in length. The number of tufts shall be as specified in table I. Mattresses shall be tufted through both sides. Knots of tufts shall be arranged so that they will not appear above the surface of either side of the mattress for outer tufted mattresses. The arrangement of the tufts shall be in a diamond pattern and spaced not more than 8 inches apart.

3.5.10.1 Innertufted mattresses. Mattresses can also be innertufted resulting in a smooth surface mattress with concealed innertufting. The tufting requirement specified in 3.5.10 shall also apply to innertufted mattresses. Innertufted mattresses may require the use of a tufting support fabric to properly anchor the filling materials. The tufting support fabric shall be as specified in 3.3.6. Any tear, rip or cut in the tufting support fabric shall be repaired prior to filling the mattress.

3.6 Physical requirements. The size, dimensions, weight and number of tufts per mattress shall be as specified in table I.

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TABLE I. Size, dimensions, weight, and tufts

Size	Dimensions (inches) Length, ± 1/2 inch	Width, ± 1/2 inch	Thickness, ± 1/2 inch	Weight, pounds (minimum)	Tufts (minimum)
1	72-1/2	26	4	18	28
2	76	26	3	15	28
3	73	27	4	28	28
4	74	28	4	25	28
5	77	28	4-1/2	23-1/2	28
6	76	30	4	23	28
7	78	31	4	27	38
8	75	33	6	37	38
9	76	34	4	26	38
10	78	36	4-1/2	39	4-1/2
11	75	38	6	42	38
12	76	39	4-1/2	38	38

3.7 Workmanship. The finished mattresses shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article sample submitted in accordance with 3.1 shall be inspected as specified in 4.4.3 for compliance with design, material, construction, workmanship, weight, and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Testing of components. In addition to the quality assurance provisions of the referenced specifications, components and materials listed in table III shall be tested for the characteristics shown therein. The methods of testing specified in FED-STD-191, wherever applicable, and as specified herein shall be followed. When the data in the "Number of determinations per sample unit" and the "Results reported as" columns are not specified in table III. they shall be as required in the referenced test method. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be in accordance with table II and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

TABLE II. Sample size

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. Test methods of components

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as	Sample unit
Edge binding tape (1 cone or spool)	Material identification	3.3.4.1	Standard commercial	1 1	Pass or fail	
	Breaking strength	3.3.4.1	6016 1/	-	-	1 cone, spool or 30 feet of tape
	Width	3.3.4.1	5020	3	Pass or fail	
Tufting tape, braid or twine (1 cone or spool)	Material identification	3.3.4.2	Standard commercial	1	Pass or fail	
	Breaking strength	3.3.4.2	6016 1/	-	-	1 cone, spool or 30 feet of tape
	Width (when applicable)	3.3.4.2	5020	3	Pass or fail	

1/ The full width of the tape shall be used.

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4.4.1.2 Components, materials and component assemblies. When the components and materials listed below are used in the fabrication of mattresses, a supplier's certificate of compliance (see 4.4) shall be furnished and will be acceptable for the specified requirements.

Component materials	Requirement paragraph
Felt 1/	3.3.1
Grades of felt (as applicable) 1, 2/	3.3.1.1 through 3.3.1.4
Cotton 1/	3.3.1.5
Linters 1/	3.3.1.6
Polyester fiber	3.3.1.7
Barrier materials	3.3.5
Inner support tufting fabric	3.3.6
Heat shield or flange fabric	3.3.7

1/ Unless otherwise specified (see 6.2), the cotton or linters or felt shall be examined, to determine conformance with 3.3.1.1 through 3.3.1.6, by classifiers licensed by the U.S. Department of Agriculture; or representative samples, each weighing not less than 1/2 pound, of the cotton or linters and a minimum 12 inches by 12 inches of felt shall be forwarded to the administrative office concerned at 4841 Summer Ave., Memphis, TN 38122, where inspection by the proper board of the U.S. Department of Agriculture shall be obtained. The opinion of this board as to the true grade or quality will be final. Inspection reports from the USDA shall be furnished and will be acceptable for the stated requirements.

2/ Not applicable to grade D felt.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated non-conformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements.

Operation or assmebly	Requirement paragraph
Construction of cover (with or without heat shield or flange)	3.5.5
Construction of boxside panel	3.5.6
Construction of edges	3.5.7
Distribution of felt padding	3.5.8
Application of barrier material (toppers) to felt (when applicable)	3.5.9
Concealed operations of innertufted mattresses	3.5.10 and 3.5.10.1

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4.4.3 Inspection of the end item.

4.4.3.1 Visual examination of mattresses. The defects found during examination shall be classified in accordance with 4.4.3.1.1, 4.4.3.1.2, and 4.4.3.1.3. The sample unit for these examinations shall be one mattress (as applicable). Any mattress removed from the polyethylene bag for examination need not be resealed. The open end of the polyethylene bag shall be repaired (taped up) prior to placing the packaged mattress (mattress in bag) back into a carton. The inspection levels and AQL's shall be as specified in 4.4.3.2.

4.4.3.1.1 Examination of mattress for defects in design, material, construction and workmanship.

Examine	Defect	Classification Major	Minor
Quality of ticking	Hole, cut, tear, smash, mend or open place:		
	-up to 1 inch, inclusive		X
	-more than 1 inch	X	
	Loose yarn, snagged thread, misweave, missing yarn, broken yarn, thin place, shade bar, or woven in waste:		
	-up to 1 inch, inclusive		X
	-more than 1 inch	X	
	Rancid or other objectionable odor	X	
	Not fabric specified	X	
	Start marks (heavy or light)		X
	Spot or stain 1/4 inch or over in diameter, clearly noticeable when viewed at a distance within 3 feet		X
	Excessive neppiness	X	
Seams and stitching	Any required seam or stitching operation omitted	X	
	Ticking not seamed with seam type SSa-1 or LSb-2		X
	Not stitch type 301 or 401 (as applicable)		X
	Stitches per inch less than specified	X	
	Stitches per inch more than specified		X
	Ends of any seam not backstitched (except when end of seam is caught in another seam or stitching, or when closing the mattress) as specified		X

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Examine	Defect	Classification Major	Classification Minor
Seams and stitching (cont'd)	Raw edge more than 1 inch in length		X
	Open seam more than 1 inch in length	X	
	Open seam 1 inch or less in length		X
	NOTE: A seam shall be classified as open when one or more stitches are broken or when two or more continuous skipped stitches or run-offs occur.		
	Stitch tension loose, resulting in a loosely secured seam		X
	Stitch tension tight (stitches break when normal strain is applied to the seam or stitching)		X
	NOTE: Puckering is evidence of tight tension, when puckering is evident, the inspector shall test seams by exerting pull on seam.		
	Needle chew, hole or cut: -up to 1 inch, inclusive		X
	-more than 1 inch	X	
	Seam badly puckered, pleated, twisted or caught in unrelated operation		X
	Five or more untrimmed thread ends or loose threads, 1 inch or more in length		X
Construction and workmanship	Sagging appearance, i.e., filling not evenly distributed	X	
	Lumps clearly noticeable	X	
	Thin hollow places clearly noticeable	X	
	Top or bottom section of mattress case fabricated of more than two lengthwise pieces		X
	Bottom section of mattress case fabricated of more than two lengthwise pieces		X
	Blue stripe of ticking does not run in lengthwise direction		X
	Top and bottom ticking, when pieced, not properly matched along piecing seam, affecting appearance		X
	Any cross seam in top and bottom ticking		X

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Examine	Defect	Classification Major	Minor
Boxside panel	Boxside panel formed of more than four separate pieces		X
	Joining seams of boxside panel (when applicable), not located at a corner		X
	Pattern not properly matched when joining seams.		X
Edge binding tape	Binding tape does not extend completely around edge of mattress	X	
	More than 4 pieces of binding tape on edge of mattress	X	
	Binding tape puckered, pleated, or twisted		X
Tufting (outer-tufted mattresses)	Material used in tufting not type specified		X
	Any broken tuft.	X	
	Three or more tufts insecurely knotted or fastened.	X	
	Two tufts insecurely knotted or fastened.		X
	Not accomplished as specified.		X
	Tufts not arranged in a diamond pattern.		X
	Two or more less than number of tufts specified.	X	
	One less than number of tufts specified		X
	Any knot appearing above surface plane on either side of mattress.		X
	Tear over 3/4 inch in ticking under tuft (check a minimum of 3 tufts on each side of mattress).		X
	Uneven overall tufting causing mattress to lean over or to have an uneven surface appearance.	X	
	Depth of tufts not as specified		X

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Examine	Defect	Classification Major	Classification Minor
Tufting (outer-tufted mattresses) (cont'd)			
	NOTE: Check a minimum of three tufts by inserting a pointed thin piece of metal or wire, and of sufficient length, through the tuft until the point is felt on the bottom side of mattress. Grasp the instrument at the beginning of the depth of the top tuft and remove. Measure the distance from point of instrument to where thumb and forefinger are holding the instrument.		
Labels	Missing, incomplete, incorrect or illegible	X	
	Not sewn in same edge seam at end of mattress		X

4.4.3.1.2 Examination of mattress for defects in dimensions.

Examine	Defect
Length (see 4.5.2)	Average length is beyond tolerances indicated in table I.
Width (see 4.5.2)	Average width is beyond tolerances indicated in table I.
Spacing of tufts (outer tufted mattresses) both surfaces	More than 8 inches apart.
Thickness (see 4.5.1)	Average thickness is beyond tolerances indicated in table I.

4.4.3.1.3 Examination of mattress for defects in weight.

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Examine	Defect
Weight	Less than the minimum specified in table I. 1/

1/ Weight. Weight may be determined under prevailing atmospheric conditions, except in the settlement of disputes. In case of dispute, weight shall then be determined under standard atmospheric conditions as defined in FED-STD-191. Deviations from the minimum specified weight requirements due to the use of grade D felt is subject to the approval of the contracting officer.

4.4.3.2 Inspection levels and acceptable quality levels (AQLs). The inspection levels and AQLs expressed in defects per 100 units shall be as follows:

Examination	Inspection level	AQL	
		Major	Major and minor combined
4.4.3.1.1	I	4.0	10.0
4.4.3.1.2	S-3	4.0 (1 class)	
4.4.3.1.3	S-3	4.0 (1 class)	

4.4.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-M-3996 to determine that packaging, packing and marking requirements comply with section 5 requirements of this specification.

4.5 Methods of measurement.

4.5.1 Measurement of thickness. Thickness shall be the average of the measurements taken at each of the four corners of the mattress. Each thickness measurement shall be made by determining the free height under the corner of an 18-inch square panel, with a total weight load of 4 pounds, resting on the mattress with two edges of the panel over the adjacent edges of the mattress.

4.5.2 Measurement of length and width. Length and width shall be taken as the average of three measurements in each direction. Before measuring, mattresses shall be shaken out after removal from containers (when applicable) and laid in a flat position.

4.6 Flammability. Flammability tests shall be performed in accordance with all the provisions of the Consumer Product Safety Commission Flammability Standard for Mattresses FF-4-72. A supplier's certificate of compliance shall be furnished and will be accepted for the stated requirements. Certification shall include construction and material details used to meet the standard.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Mattresses shall be packaged in accordance with the applicable requirements of MIL-M-3996.

5.1.2 For civil agency procurements, where level B packaging is specified in the contract or order, the requirements in 5.1.1 (level A) shall apply.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Mattresses shall be packed in accordance with the applicable requirements of MIL-M-3996.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123.

5.3.1.1 Special marking. The following marking shall also appear on at least two side panels of each shipping container in letters not less than 1-1/2 inches in height.

STOW FLAT-DO STOW ON END OR SIDE

5.3.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-M-3996.he preparing act

6. NOTES

6.1 Intended use. The mattresses are intended for use by Civilian and Military agencies of the Government.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Grade, and size mattress required (see 1.2 and 3.6).
(c) When a first article sample is required (see 3.1).
(d) When fire-resistant ticking is required (see 3.3.2).

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(e) When medical label is required (see 3.3.8.3).
(f) When examination of cotton or linters or felt is not required to be examined by the USDA (see 4.4.1.1).
(g) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(h) When grade D felt is required (see 3.3.1.4)

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one finished mattress. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Flame retardant and smolder resistant finishes and materials. Approval of such components and combinations is the responsibility of the U. S. Army Natick Laboratories, Natick, MA 01760, and is based on extensive tests including those for toxicity performed by Office of the Surgeon General which are not set forth in this specification. Because of the time necessary to conduct full evaluation, only those chemical treatments and materials already approved and so listed in the invitation for bid or request for proposal shall be considered acceptable for the related procurement.

6.5 Bonding agents. Approval of bonding agents for use in battings covered by this specification, is the responsibility of the U. S. Army Natick Laboratories, Natick, MA 01760. Approval is based on more extensive evaluations, including those of toxicity, which are not set forth in the specification. Because of the time necessary to conduct full evaluation, only those bonding agents already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

6.5.1 Submittal for approval. Request for approval should be submitted to the U. S. Army Natick Laboratories, attention of AMXNM-TDO. It should be accompanied by approximately five square yards of a minimum of 2 ounce/square yard polyester batting manufactured with the proposed bonding agent together with applicable test data. Information is to be furnished regarding the commercial designation and chemical description of the bonding agent. (Approved lists will indicate only the commercial designation). Chemical identification with products already approved should be made, if applicable, and any available data on skin irritancy or toxicity should be submitted. If applicable, a statement should be included that as used the resin is well cured and that no materials with known high toxicity are used prior to cure.

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6.6 Supersession data. This specification includes the requirements of Interim Purchase Description IP/DES-S-4-3 dated 3 May 1973. Types I, II and III mattresses have been replaced by the taped box edge mattress covered by this specification.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 82

Review activities:

Army - MD
Navy - MC
Air Force - 85
Defense Supply Agency - DM

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

AGR - AMS
COM-NBS
DC GOV'T
GSA-FSS
HEW-FDA
INT - BPA
JUS-FPI
VA - DMS

Project No. 7210-0121

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